

CUSTOMER BEHAVIOUR ANALYSIS PROJECT REPORT

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1. Introduction

Customer Behaviour Analysis helps organizations understand purchasing patterns, customer preferences, spending habits and customer loyalty. This project analyzes 3,900 retail transactions using Python, PostgreSQL and Power BI to generate business insights and support data-driven decision making.

2. Business Problem

Retail businesses generate huge amounts of customer data but often fail to transform it into actionable insights. The company wants to identify top-selling products, customer segments, subscription behaviour, preferred shipping methods and purchasing trends in order to increase revenue and customer satisfaction.

3. Business Objectives

- Clean and preprocess the dataset.
- Handle missing values and duplicates.
- Perform exploratory data analysis.
- Solve business questions using SQL.
- Build an interactive Power BI dashboard.
- Generate actionable business recommendations.

4. Dataset Description

The dataset contains 3,900 customer records and 18 attributes including Age, Gender, Product Category, Purchase Amount, Review Rating, Subscription Status, Shipping Type, Previous Purchases, Payment Method and Purchase Frequency.

5. Data Cleaning & Preprocessing

The Review Rating column contained 37 missing values. These missing values were replaced with the mean value to preserve the dataset. Duplicate records were checked and no duplicates were found. Unique values of all categorical columns were verified before analysis.

6. Exploratory Data Analysis

EDA was performed to understand customer demographics, purchasing behaviour, payment preferences, product popularity and purchase frequency. Descriptive statistics and frequency distributions were calculated. Customers were also grouped into age segments for further analysis.

7. SQL Business Analysis

Business questions were answered using SQL aggregation functions, CASE statements, GROUP BY, ORDER BY, window functions (DENSE_RANK), filtering and ranking. Analyses included top-rated products, shipping comparison, subscription revenue, discount analysis, customer segmentation, top products by category and repeat buyer analysis.

8. Data Visualization

Interactive dashboards were created in Power BI using KPI cards, bar charts, donut charts, slicers and column charts. Users can filter data by category, gender, shipping type and subscription status for interactive analysis.

9. Key Business Insights

Clothing generated the highest revenue. Adult customers represented the largest customer segment. Average purchase amount was approximately \$60 and average review rating was 3.75. Loyal customers formed the largest customer segment. Express shipping customers spent slightly more on average. Jewelry, Blouse and Pants were among the highest-selling products.

10. Business Recommendations

Increase inventory for high-performing categories, strengthen loyalty programs, improve subscription benefits, promote highly rated products, provide personalized offers to returning customers and optimize shipping options based on customer preferences.

11. Conclusion

This project demonstrates a complete business analytics workflow using Python, PostgreSQL and Power BI. Data cleaning, EDA, SQL analytics and dashboard development were combined to transform raw customer data into meaningful business insights.

12. Tools & Technologies

Python, Pandas, NumPy, PostgreSQL, SQL, Power BI, Microsoft Excel, Data Cleaning, EDA, Dashboard Design, Business Intelligence and Data Visualization.